



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-11

Experiment 11. Tic Tac Toe Game

Aim

To implement a Tic Tac Toe game using Python.

Objectives

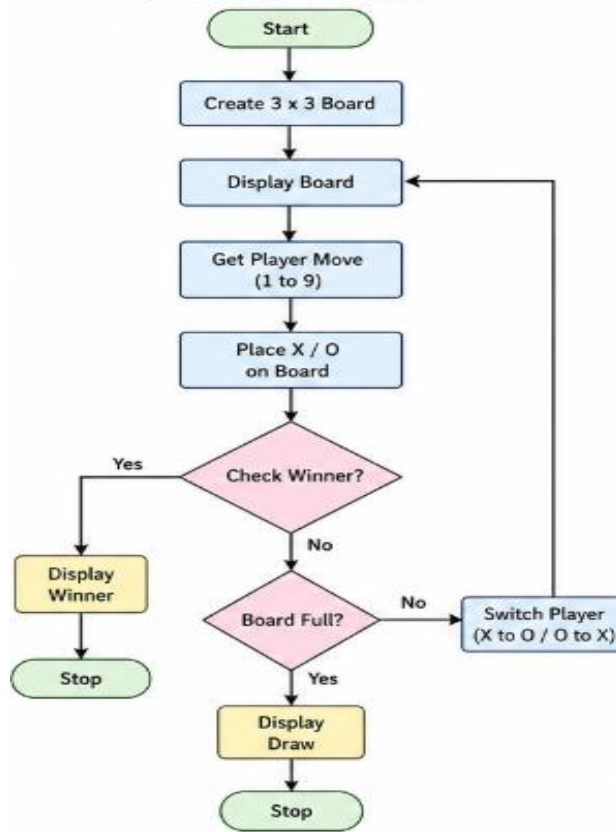
- To understand basic game logic.
- To implement a 3×3 game board.
- To handle player turns.
- To identify the winner or draw.

Algorithm

1. Create a 3×3 board.
2. Get the player's position.
3. Place X or O.
4. Check the winning combinations.
5. Display the winner or draw.
6. Switch the player and repeat.

Flowchart

11. Tic Tac Toe Game



Code

```
board = [" "] * 9
```

```
def win(p):
```

```
    for a, b, c in
```

```
        [(0,1,2),(3,4,5),(6,7,8),
```

```
         (0,3,6),(1,4,7),(2,5,8),
```

```
         (0,4,8),(2,4,6)]:
```

```
        if board[a] == board[b] ==
```

```
board[c] == p: return True
```

```

        return False

p = "X"

for i in range(9):

    pos = int(input(f"{p} position
(1-9): ")) - 1

    if board[pos] == " ":

        board[pos] = p

        if win(p):

            print(p, "Wins!")

            break

        p = "O" if p == "X" else "X"

else:

    print("Draw")

```

Output

The screenshot displays a Python IDE with two windows. The left window shows the source code for a Tic Tac Toe game. The right window shows the output of the program, which is a series of moves by player X, resulting in a win.

```

ai exp 11.py - C:/Users/shrut/AppData/Local/Programs/Python/Python314/ai exp 11.py (3.14.2)
File Edit Format Run Options Window Help
board = [" "] * 9

def win(p):
    for a, b, c in [(0,1,2), (3,4,5), (6,7,8),
                    (0,3,6), (1,4,7), (2,5,8),
                    (0,4,8), (2,4,6)]:
        if board[a] == board[b] == board[c] == p:
            return True
    return False

p = "X"

for i in range(9):
    pos = int(input(f"{p} position (1-9): ")) - 1

    if board[pos] == " ":
        board[pos] = p

        if win(p):
            print(p, "Wins!")
            break

        p = "O" if p == "X" else "X"

else:
    print("Draw")

```

```

Python 3.14.2 (tags/v3.14.2:df
on win32
Enter "help" below or click "H
>>>
= RESTART: C:/Users/shrut/AppDa
X position (1-9): 1
O position (1-9): 2
X position (1-9): 3
O position (1-9): 3
O position (1-9): 4
X position (1-9): 5
O position (1-9): 5
O position (1-9): 6
X position (1-9): 7
X Wins!
>>>

```

Result

The program successfully executes the Tic Tac Toe game and displays the winner or draw.

Conclusion

Thus, the Tic Tac Toe game was successfully implemented using Python.